INCH-POUND

A-A-58088A
14 MAY 2003
SUPERSEDING
A-A-58088
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COMMERCIAL ITEM DESCRIPTION
SCALE, WEIGHING SYSTEMS, AIRCRAFT PLATFORM

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers 3 types of portable, battery powered platform weighing systems suitable for weighing aircraft in the U.S. Air Force inventory. The system consists of platform scales and accessories as specified in paragraph 3.2. Multiple systems may be required for some aircraft.

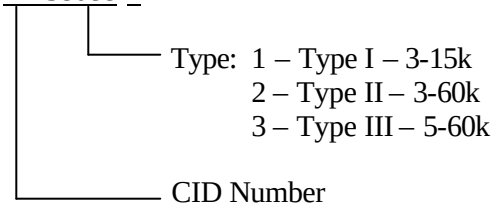
2. CLASSIFICATION.

2.1 The weighing systems shall be of the following types:

Type I: 3-15k (3 platform scales, 15,000 lb. capacity each)
Type II: 3-60k (3 platform scales, 60,000 lb. capacity each)
Type III: 5-60k (5 platform scales, 60,000 lb. capacity each)

2.2 Part or Identifying Number (PIN). The following part or identifying numbering procedure is for government purposes and does not constitute a requirement for the contractor.

Example of reference part number: AA58088-1



PIN AA58088-1 would identify a Type I Aircraft Weighing System with a 15,000 lb. platform scale.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Warner Robins Air Logistics Center, WR-ALC/TILCC, Robins AFB, GA 31098-1640.

AMSC N/A FSC 6670

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3. SALIENT CHARACTERISTICS. The following are minimum requirements, unless otherwise indicated. Information in parenthesis is for clarification or guidance only; it is not a requirement.

3.1 General. The weighing system shall be easy to use, maintain, and repair. It shall be durable, reliable, and free of defects affecting performance, strength, and safety. It shall be complete, including all items required to weigh aircraft except tow tug and aircraft peculiar tooling. Service life shall be at least 15 years under normal Air Force operating and storage conditions and environments.

3.2 Components. The platform weighing systems shall include the following major items and additional items as required.

TABLE I

ITEM	TYPE I QUANTITY	TYPE II QUANTITY	TYPE III QUANTITY	PARAGRAPH
Platform Scales	3	3	5	3.3
Power Cords	3	3	5	3.3.5
Entry Ramps	3	3	5	3.5
End Stops	3	3	5	3.6
Bridge	0	1	2	3.7
Trailer	1	1	1	3.8
Accessory Kit	1	1	1	3.9

3.3 Platform scale. The scale shall be self-contained and battery powered, with an integral digital display. When platforms are side by side, the distance between weighing areas shall not exceed 10 inches. The scale shall withstand a 50% overload with no damage or change in calibration. Overall dimensions shall not exceed 40 inches long by 40 inches wide excluding display.

3.3.1 Type I: 3-15k. 0 to 15,000 lb. weighing range. Accuracy shall be within +/- 0.10% of the applied load or +/- 4 lb., whichever is greater. The weighing area shall be a minimum of 15 inches wide by 15 inches long. The height of the weighing area shall not exceed 1.75 inches, so that separate ramps, bridges and stops are not required. Weight shall not exceed 37 lb.

3.3.2 Type II: 3-60k and Type III: 5-60k. 0 to 60,000 lb. weighing range. Accuracy shall be within +/- 0.10% of the applied load or +/-10 lb., whichever is greater. Weighing area shall be a minimum of 22 inches wide by 30 inches long. The height of the weighing area, excluding end stop, shall not exceed 4 inches. Ramp, bridge, and stop connections shall be identical. Weight shall not exceed 150 lb. The load cell(s) used within the platform scale shall be commercially available and National Type Evaluation Program (NTEP) certified to 10,000 divisions NIST/NTEP Class IIIL by the National Institute of Standards and Technology (NIST).

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3.3.3 Range and accuracy. The scale shall weigh accurately and repeatably. The allowable error includes all error effects, including calibration uncertainty, display increment rounding, floor slope, platform loading errors (such as off-center loading, within the weighing area), creep over 30 minutes, temperature variation of +/-5°F, non-linearity, repeatability, zero shift, drift, etc. Range and accuracy, weighing area, if applicable, and warranty expiration date shall be marked on the scale.

3.3.3.1 Floor slope. The scale shall measure accurately when placed longitudinally or laterally on hangar floors, or flight line areas, with slopes up to +/-2.1% (1.2 degrees), without platform shimming or leveling.

3.3.3.2 Gravitational correction factor. The scale shall be calibrated to standard gravity (acceleration due to gravity shall be equal to 32.174 ft/sec² at 45 degrees latitude and sea level.) When the scale is turned on, the stored latitude and elevation shall be displayed and the user shall have an opportunity to change the values. The indicated weight shall be automatically adjusted to standard gravity. (Table 1 provides correction factors that may be programmed into the scale.) The scale shall be marked to indicate that the weight has been adjusted to standard gravity.

3.3.4 Display. The display shall be easy to read from a standing position in either dark or direct sunlight conditions (leading zeros blanked, proper display increments, illuminated display, and contrast adjustment). It shall include a self test, zero calibration button, low battery indicator, automatic shut-off and temperature compensation feature. A means to input latitude and elevation shall be provided. The indicator element of the display shall be NTEP certified to 10,000 divisions by NIST.

3.3.5 Power. The scale shall operate from 120/220/240 VAC, 50/60Hz with fifty (50) foot grounded power cord with NEMA 5-15 configuration plug which shall be used to power the scale and/or recharge the batteries.

3.3.5.1 The scale shall operate off commercially available alkaline or Ni-Cad or equivalent rechargeable batteries pack for at least 20 hours of continuous use between charges with an indicator for low battery power on the display.

3.4 Calibration. The scale shall be easily calibrated and the calibration protected against accidental destruction. The user shall not be required to use calibration curves. Calibration procedures and equipment shall be in accordance with ANSI/NCSL Z540-1, sections 7 through 18. If the force standard uncertainty exceeds 25 percent of the tolerance, the calibration tolerance shall take the force standard uncertainty into account. (Example: If the standard uncertainty at 10,000 lb. is ±4 lb., then the scale indication shall be within ±6 lb.) A calibration tag shall be placed near the display. Calibration certificates or reports and any special tooling or equipment required for calibration shall be supplied with the weighing system, except press, force standard, load distribution plate(s) and standard hand tools. The scale shall indicate within tolerance over a 1 year calibration interval.

3.5 Entry Ramp. The ramp shall be easily attached and have a non-skid top surface. The ramp slope shall not exceed 3°.

3.6 End Stop. The stop shall have sufficient height to prevent the aircraft from rolling off the platform and be easily attached to the platform.

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3.7 Bridge. The bridge shall be easily attached to the platforms and have non-skid top surface. The length of the bridge shall be such that when platforms are assembled for tandem weighing, the distance between the platform weighing areas adjoining the bridge is 29 inches or less and the distance between the platform weighing areas adjoining the ramp and stop is a minimum of 85 inches.

3.8 Storage and transportation trailer. Four wheeled trailer suitable for easy handling and safely towing the weighing system, at speeds up to 10 mph and over 1 inch bumps, in and out of hangars, around flight line areas and long term storage in an aircraft hangar. It shall provide environmental protection for towing outside (minimum: a durable waterproof cover.) It shall have a detachable, hinged tow bar with eye in accordance with A-A-52464. It shall have a rear pintle hook in accordance with A-A-52550, suitable for towing 2 fully loaded trailers. It shall have parking brake, fork lift provision, and tie-down rings. The trailer shall be marked with the following: maximum towing speed, lubrication information, tire pressure, and if applicable, "STORE INSIDE". The identification tag shall have appropriate block for the warranty expiration date. All bolts with nuts shall have at least 2 complete threads protruding past the nut.

3.9 Accessory kit. The accessory kit, as a minimum, shall consist of the following items:

TABLE II

QTY	ITEM	DESCRIPTION
1	Leveling bar	3/8 x7/8 x 37 inches long. Parallel and straight within 0.003 inches, corrosion resistant, 1 or 2 parts.
1	Level	Suitable digital protractor with 0.01 deg resolution (See 3.9.1 below.)
2	Plumb bob	3-½ ounce, corrosion resistant, 15 foot cord
1	Marking Chalk	For marking points on hangar floor
1	Ruler	12 inch, stainless steel, graduated in 0.10 inches
1	Chalk line reel	100 foot, with chalk
1	Steel tape	50 foot, with reel, marked and numbered to 600 inches, graduated in 0.10 inches
1	Fuel dipper	For taking fuel samples from aircraft fuel tanks, includes chain and handle, corrosion resistant
2	Hydrometer	Covers 5.5 to 7.5 lb. per U.S. gallon
1	Hydrometer jar	Plastic cylinder with base, for use with hydrometer
1	Tools	As needed to assemble leveling bar
1	Case	Plastic with durable hinges and latches. Attached cushioning material protects items. Top marked

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3.9.1 Level Specification.

3.9.1.1 Design. The level shall consist essentially of a stock, adjustable level vial, vial protector, fixed plumb vial, and cross test vial.

3.9.1.2 Accuracy. The level accuracy is the measure of the allowable deviation or tolerance on the level reading itself. It is determined by the setting of the vial within the stock as checked against a true horizontal or vertical plane.

3.9.1.3 Level Length. The level length shall be 12 inches.

3.9.1.4 Level Base. The level width of the base shall be 1 3/8 inches minimum to 1¾ inches maximum.

3.9.1.5 Level Weight. The weight of the level shall be no greater than 5 lbs. maximum.

3.9.1.6 Vials. The vials provided shall be a lime or heat-resistant boro-silicate glass tube and shall be ground to a vial sensitivity of 90 seconds maximum. There shall be no less than two (2) auxiliary lines, permanently marked, spaced 1/10 inch apart and 1/10 inch from the outside of both bubble defining lines, and the value of these lines shall be permanently marked on the level. The plumb and cross test vial shall be fixed in the stock and shall be commercial bent-glass tubes and shall be permanently marked with two (2) black bubble-defining centering lines. The vials shall be bent to a radius of 7.645 inches maximum. The radius of curvature shall be uniform along the length of the vial body. Alternatives to these vials shall be considered as long as specified tolerances for sensitivity and accuracy are met and/or exceeded.

3.9.1.6.1 Vial Sensitivity. Measurement of vial sensitivity is the amount of angular movement in a vertical plane required to move the vial bubble 1/10 inch. Angular movement may be expressed in degrees, minutes, or seconds. The sensitivity of the bubble in the motion in the vial is governed by the vial radius; the larger the radius, the smaller the angle required to move the bubble 1/10 inch.

3.9.1.6.2 Vial Accuracy. The main level vial shall be adjustable and shall be accurate for the main vial setting not to exceed 0.001 inches/foot. The level concavity shall not exceed a tolerance of 0.002 inches.

3.9.1.6.3 Vial Protector. The holder for the main level shall be equipped with means for protecting the vial from damage. This device shall be readily accessible and easily operated.

3.9.1.7 Material/Base. The base material required shall be cast iron or similar material designed not to warp, properly support vials or vial holders, and shall be within the specified tolerances. The base shall be provided with a vee or concave groove through its entire length. Sufficient flat margin on each side of the groove shall be provided for a good seat for flat work. The groove shall be straight and shall provide a true seat for cylindrical work.

3.10 Assembly. The weighing systems shall be assembled for normal weighing, or disassembled, by 2 people using only the items in the weighing kit. Type I: 3-15k and Type II: 3-60k shall be assembled in 10 minutes and Type III: 5-60k in 15 minutes.

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3.11 Marking. Identify weighing system in accordance with MIL-STD-130, including NSN, contract number, part or identifying number and warranty information. Any item weighing more than 37 lb. shall be marked with the weight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Market acceptability. The following market acceptability criteria is necessary to document the quality of the product to be provided under this CID. The offered platform scale must have been on the market for at least 2 years. One hundred of the offered scales must have been sold in the commercial market place.

4.3 Product sample. A product sample may be required, for examination, testing and service testing with an aircraft to determine whether it provides acceptable form, fit, function and suitability for use. If a product sample is required, shipping instructions will be provided.

4.3 Acceptance testing. After calibration, each platform shall be acceptance tested at a minimum of 5 loads covering the weighing range. Any scale that indicates out of tolerance shall be rejected and shall have documented corrective action and procuring agency approval before being resubmitted for acceptance testing. Re-calibration alone is not acceptable corrective action.

5. PACKAGING.

Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 ANSI/NCSL Z540-1. "Calibration Laboratories and Measuring and Test Equipment – General Requirements" may be obtained from National Conference of Standards Laboratories, 1800 30th St., Suite 305B, Boulder, CO 80301, (303)440-3339 or American National Standards Institute (ANSI), 11 W 42nd St., 13th Floor, New York, NY 10036, (212) 642-4900.

6.2.2 ASTM E74. "Standard Practice of Calibration of Force-Measuring Instruments for Verifying the Force Indication of Testing Machines" may be obtained from American Society for Testing and Materials, 100 Barr Harbor Dr., West Conshohocken, PA 19428-2959, (610) 832-9500.

6.2.3 NIST/NTEP Class IIIL. Specifications, tolerances, and other technical requirements for weighing and measuring devices, NIST Handbook 44, 2003 edition may be ordered from U.S. Government Printing Office, Stop SSOP, Washington, D.C. 20402-0001, Telephone (202)512- 1800 or via internet at www.nist.gov (Publications).

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6.2.4 Other. MIL-STD-130 “Identification Marking of U.S. Military Property”, A-A-52550 “Pintle Assembly, Towing, 18000, 40.000, and 100,000 lbs. Capacity”; and A-A-52464 “Coupler, Drawbar, Ring: Light Duty, 60,000 lb. GVW; Offset (Taper Shank), 60,000 lb. GVW; and Heavy Duty 120,000 lb. GVW” may be obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094, (215) 697-1187, FAX (215) 697-2978, or Customer Service (215) 697-2667/2179.

6.3 Ordering data.

a. Identify Type required (paragraph 3.2).

b. Identify whether a product sample is required (paragraph 4.3).

6.4 Commercial availability. Commercially available is defined as availability from three or more sources as a standard cataloged item.

MILITARY INTEREST:

Custodian:
Air Force – 99

Preparing Activity:
Air Force – 84

Agent:
Air Force – 99

(Project Number 6670-0217)

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GRAVITATIONAL CORRECTION FACTORS
FOR LATITUDE AND ELEVATION

Weighing System is calibrated to standard gravity
G=980.665 cm/sec² (32.174 ft/sec²) at 45 degrees latitude and sea level

TABLE III
CORRECTION FACTORS

LATITUDE	ELEVATION ABOVE SEA LEVEL							
(DEG)	0	1000	2000	3000	4000	5000	6000	7000
0	1.0027	1.0028	1.0029	1.0030	1.0031	1.0032	1.0033	1.0034
5	1.0026	1.0027	1.0028	1.0029	1.0030	1.0031	1.0032	1.0033
10	1.0025	1.0026	1.0027	1.0028	1.0029	1.0030	1.0031	1.0032
15	1.0023	1.0024	1.0025	1.0026	1.0027	1.0028	1.0029	1.0030
20	1.0021	1.0022	1.0023	1.0024	1.0025	1.0025	1.0026	1.0027
25	1.0017	1.0018	1.0019	1.0020	1.0021	1.0022	1.0023	1.0024
30	1.0014	1.0015	1.0016	1.0017	1.0017	1.0018	1.0019	1.0020
35	1.0009	1.0010	1.0011	1.0012	1.0013	1.0014	1.0015	1.0016
40	1.0005	1.0006	1.0007	1.0008	1.0009	1.0010	1.0011	1.0012
45	1.0000	1.0001	1.0002	1.0003	1.0004	1.0005	1.0006	1.0007
50	0.9996	0.9997	0.9998	0.9999	1.0000	1.0001	1.0002	1.0003
55	0.9991	0.9992	0.9993	0.9994	0.9995	0.9996	0.9997	1.9998
60	0.9987	0.9988	0.9989	0.9990	0.9991	0.9992	0.9993	0.9994
65	0.9983	0.9984	0.9985	0.9986	0.9987	0.9988	0.9989	0.9990
70	0.9980	0.9981	0.9982	0.9983	0.9984	0.9985	0.9986	0.9987
75	0.9978	0.9979	0.9979	0.9980	0.9981	0.9982	0.9983	0.9984
80	0.9976	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981	0.9982
85	0.9974	0.9975	0.9976	0.9977	0.9978	0.9979	0.9980	0.9981
90	0.9974	0.9975	0.9976	0.9977	0.9978	0.9979	0.9980	0.9981

INSTRUCTION: To obtain weight adjusted to standard gravity:

(1) Determine the latitude and elevation where the scale is being used.
(2) Round to the nearest latitude and elevation shown on the chart.
(3) Multiply the weight indicated on the scale by the correction factor shown.

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APPENDIX

(This is provided for informational purposes only.)

10.1 Force standard. The force calibration press shall be a primary or secondary standard calibrated in accordance with ASTM E74. (Example: the uncertainty of the standard used to calibrate the platform scale at 10,000 lb. shall not exceed ± 5lb.) The presses used by the Air Force Precision Measuring Equipment Laboratory (PMEL) have a maximum length and width limitation of 40 inches.

10.1.1 If the force standard uncertainty exceeds 25% of the tolerance, the calibration tolerance shall take the force standard uncertainty into account. (Example: If the standard uncertainty at 10,000 lb. is ±4 lb., then the scale indication shall be within ±6 lb.)

10.2 Calibration. The Air Force PMELs currently use HBM load cells, P/Ns C3H3-10T/AF and C3H3-30T/AF with GEC indicator P/N LCI-00000-1 as secondary standards. The load cells have nominal loads of 20,000 and 60,000 lb., and are calibrated in accordance with ASTM E74.

10.3 Uncertainty. The uncertainty, as described in ASTM E74-91 Section 8.4, shall not exceed 50% of the acceptable tolerance. The load cells have uncertainty within ±0.017 percent of nominal load; ±0.02 percent, including indicator error.

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